Speaker Notes: Slides 1 to 4

Vehicle Loan Risk Check · IT3051 Fundamentals of Data Mining · Mini Project 2026

How to use these notes

- Total speaking time: about **4 to 4.5 minutes** for the four slides.
- The **Script** is what to say, in plain spoken English. You do not need to read it word for word; the words in **bold** are the parts to get right.
- **Keep in mind** is background for you, useful if someone asks a question.
- Speak slowly when you say the numbers, and point to them on the slide.

Slide 1 · Vehicle Loan Risk Check (title)

Time: about 30 seconds | Purpose: Introduce the group and the project, and tell the audience what is coming.

SCRIPT

Good morning, everyone. We are Group ____, and our project is **Vehicle Loan Risk Check**.

In one sentence: it is a tool that helps a lender **spot risky vehicle loans before the money is paid out**.

I will start by explaining the problem, how our tool works, and what we learned from the data. Then my teammates will walk you through the model, the results, a live demo, and how we checked fairness.

KEEP IN MIND

- Fill in your group number before presenting (the slide still shows "Group ____").
- Optionally introduce each team member by name and their part.

Slide 2 · Unpaid loans leave warning signs in the application

Time: about 1 minute | Purpose: Explain the business problem and the size of the data.

SCRIPT

Every lender faces the same problem: some borrowers do not pay back. In the data we studied, **8 in every 100** past vehicle-loan applicants did not repay.

We worked with **121,856 past applications**, and for every one of them we know the outcome: whether the person repaid or defaulted. That is what lets us learn from the past.

Each application has **38 pieces of information**: things like age, income, loan amount, employment and credit-bureau scores.

So here is the challenge. Checking every application by hand, with all 38 details, is **slow**. And two officers looking at similar applicants can reach **different conclusions**. Our goal was to give **every application a consistent, evidence-based risk check before the loan is approved**.

KEEP IN MIND

- 8 in 100 is small, which means the data is imbalanced: most applicants repay. This is why simple accuracy is misleading (a model that says "everyone repays" would be 92% accurate) and why we judged models on how well they rank risk.
- Keep the title in mind: the warning signs are already in the application, we just need to read them consistently.

Slide 3 · Our tool gives every application a risk check in seconds

Time: about 1 minute | Purpose: Show the solution in four simple steps, and stress that a person decides.

SCRIPT

This is how the tool works, in four steps.

One: the loan officer types the applicant's details into a simple web form.

Two: the tool compares this application with about **86,000 past applicants** whose outcome we know. That is our training data, after cleaning and after keeping a separate set aside for the final test.

Three: the officer gets a clear result: **Low, Medium or High risk**, the main reasons behind it, and a suggested next step.

Four: the officer decides. The tool only flags applications for a closer look. **It never refuses anyone.**

So this is **decision support, not decision making**. A person stays responsible for every lending decision.

KEEP IN MIND

- Where 86,000 comes from: 121,856 applications, minus about 14,600 duplicates, minus 21,440 kept aside for the final test, leaves roughly 85,800 for training.
- The demo later (slide 10) shows these same four steps live, so you can say "you will see this in the demo".

Slide 4 · What 121,856 past applications taught us

Time: about 1.5 minutes | Purpose: Share the four key findings from exploring and cleaning the data, then hand over.

SCRIPT

Before building any model, we explored the data. Four things stood out.

First, credit scores are the strongest warning sign. Applicants in the lowest credit-score group defaulted about **22 times in 100**. In the highest group it was about **2 in 100**. That is roughly a ten-times difference.

Second, missing information matters. When the credit-bureau details were blank, applicants defaulted more often. So we did not just fill in blanks and forget them: we treated a blank as a signal in itself.

Third, the data needed careful cleaning. We found and removed about **14,600 duplicate and near-duplicate applications**. If we had left them in, the same applicant could appear in both

training and testing, and the results would look better than they really are.

Fourth, risk differs between groups. In the past data, younger applicants and women defaulted more often. That raises a fairness question, so we checked it carefully later, and my teammate will cover that.

With these findings in hand, we moved on to building and comparing models. I will now hand over to [teammate's name], who will explain which method worked best.

KEEP IN MIND

- The duplicates point is a strong one to say confidently: it shows you avoided an optimistic, leaky evaluation.
- The fairness results are on slide 12, so only raise the question here; do not go into the numbers.
- Replace [teammate's name] with the person presenting slide 5.

Likely questions on slides 1 to 4

Q: Why not just use accuracy?

Only 8 in 100 default, so a model that predicts "everyone repays" would be 92% accurate and useless. We measured how well each model ranks defaulters above good payers (ROC-AUC) instead.

Q: What are the external credit scores?

Scores from outside credit bureaus included in the data, scaled from 0 to 1. Lower means riskier.

Q: What is a near-duplicate?

Applications that match on almost every detail, effectively the same applicant entered more than once. Keeping them lets the model "memorise" people it later gets tested on.

Q: Why 86,000 and not 121,856?

After removing about 14,600 duplicates, we held back 21,440 applications for the final test, so the model never saw them. About 86,000 remain for training.

Q: Does the tool decide who gets a loan?

No. It only flags applications for review. An officer makes every decision.

Q: If women and younger people defaulted more, is the tool biased?

We checked this directly; my teammate covers it on the fairness slide. In short, a flag means about the same default risk in every group, and a version without gender, marital status and age narrowed the gap at almost no cost in accuracy.